

Web Service Technologies (ICT 343-3)

Assignment 1

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Sustainable Operation and Continuous Development of a Heterogeneous, Distributed Web Service Portal for Urban Community Engagement

The rapidly increasing availability of on-line digital public services has resulted in local governments establishing multiple independently operated online platforms to improve social welfare, promote civic engagement, and encourage community development. Citizens often have difficulty accessing timely, accurate information about these programs due to the fragmentation of available information. One proposed solution to this issue is the creation of a heterogeneous, distributed service-oriented website that integrates a variety of local government functions into a cohesive, single-user interface. In addition to enhancing access to community-oriented service offerings, such a website could facilitate increased user interaction, greater accessibility, and establish stronger connections between local government entities and the people of the community. However, establishing a working website is not the only requirement for the success of this project. Continued operation, maintenance, and enhancement of the website, coupled with adequate funding and risk management will be critical to the sustainability of the system. The article presents an analysis of the operational, managerial, and financial requirements necessary to successfully maintain such a portal and provides insights derived from the experience of developing and implementing this idea.

To create a distributed platform bringing together different services offered by a number of authorities, there needs to be a coordinated approach to all the various kinds of technology differences that are in use today. For example, some local authorities have different data formats, different communication protocols and different types of security, while others may rely heavily on outdated technologies or provide more modern service interfaces. To manage this technology divergence, significant investment was required to provide middleware solutions such as Application Programmer Gateways and modular service applications capable of translating and mediating data. The microservices architectural model used in this project offered even greater flexibility than normal because each of the community services could operate independently. As a result of this independence, the entire platform could grow as demand increased and could be updated without affecting other parts of the system.

Unfortunately, the advantage of using microservices architecture is dependent on the ability to implement sophisticated deployment methods and provide efficient monitoring capabilities. Developing a portal of this type highlighted the importance of having a systematic approach to filing documents, as well as maintaining an active and continuous dialogue with each of the numerous authorities that provide data to the portal.

One of the major considerations was to provide good speed and scalability since it's critical for a community portal to adequately handle sudden spikes in user activity, especially during emergencies. Hosting the platform in a flexible cloud environment provided many operational benefits, including the ability to increase/decrease capacity based on current user activity (i.e., through elastic resource allocation) and use of container orchestration to efficiently distribute tasks within the application. However, testing revealed that several agencies' authority services were

unable to keep pace due to their outdated system architecture. Therefore, asynchronous processing methods were created to enable the rapid response to requests from users and maintain a high level of user service without long delays from waiting for service responses, thereby eliminating the possibility of creating negative user experiences. Therefore, these changes allowed the gateway to continue functioning even if the external services were not able to provide the expected services within the anticipated timeframe.

Security and privacy of the data will be important aspects of the implementation because the platform processes sensitive information, such as reports made by citizens and user markings used for authentication. As a result, a secure communications architecture was needed to protect the data from malicious attacks. Some of the security methods associated with the implementation of these components include; token-based authentication, encrypted data transfer, and limited access privileges for Admin functions. Best practices in the field of computer security and recognized industry standards for security have also played a role in how these methods were implemented. Token-based authentication allowed for easy verification when working with remote services, however it is important to consider how the timeframes for expiration will be managed. Privacy laws were also a factor in determining how certain design choices would be made. The collection of sensitive data, including user location information, was only intended to collect and keep as much of the sensitive information for the shortest amount of time possible, and that is how to remain compliant with the law and maintain public trust.

Interoperability has always been a foundational aspect of the project's development, both during the initial phase and as part of its long-term sustainability. The majority of the services provided by multiple agencies did not have complete documentation and had inconsistent formats. As such, wrapper services had to be created to convert all incoming data into a format compatible with the platform, and common formats were created for describing services. This improved the reliability of the services and allowed for the introduction of new services without the retrofitting of existing components, although it involved greater effort. The result of the project demonstrated that the success of the project depended upon the clarity of communication and documentation of the interactions between the parties.

Continuous effort is necessary to keep the portal operating correctly. With a distributed system, upgrades to the software, patches for vulnerabilities, and enhancements to features are all part of the process. The system could also be affected by unanticipated changes to third-party services. In order to address these issues, we utilized automation in our development/testing/deployment processes. This enabled us to implement updates in a timely and secure manner. Planning for maintenance began at the very beginning of development. In addition, we felt that documenting every detail was critical to eliminating technical debt and providing a means for future developers to maintain continuity through the printed word. Service interfaces, data structure documentation, and architectural decisions all contributed to preventing the accumulation of technical debt and ensuring a clear view of how everything works together.

The continuous tracking of the system was essential to ensure reliability. A variety of reasons including, but not limited to, network delays, security occurrences, or an external outage may result

in system failure of any distributed system. By adding monitoring tools, the ability to detect anomalies and notify users, allowed for the timely resolution of user issues before they became disruptive to a large number of users. As testing continued to identify the need for a clear process on how to deal with incidents, it was apparent that this would be necessary in building a successful product.

Any long-term viability of the portal would depend upon constant evolution. As technology continually evolves, so too do the requirements of the community. For example, feedback received from portal users demonstrated a strong desire for real-time notifications and additional languages for support. This resulted in changes to design goals and further impacted the product roadmap development. Additionally, as part of the continuing evolution of the portal, continually adding advanced technologies such as artificial intelligence to provide refined recommendations or advanced query methods would also become available. Modularly developed products allow for additional products to be built on top of a base application without having to change significantly.

In order for the Portal project to be successful over the long term, it needs to be financially sustainable. The funding provided by Government for the Portal initially would be appropriate because it supports an element of the public service. This would be a very appropriate project to receive funding from Government because many jurisdictions around the world want to fund supporting initiatives for digital transformation in order to provide more accessible and efficient services to their citizens. However, solely depending on Government funding to support this project may deter future interest by Government as priorities evolve. Therefore, it is imperative to explore alternative revenue opportunities. While most citizens will continue to have access to fundamental services through this project at no charge, Community-Based Partner Organizations and NGOs can pay to subscribe to access enhanced capabilities, specifically the analytics dashboard and service promotion features.

The ethical regulation of advertisements within the platform provides another benefit to the platform; users will not be mistrustful of this form of advertising. One example of this could be to advertise eco-friendly products, or to partner with local businesses for sponsorships. The design includes designated visual areas on the platform for this type of content, so as not to interfere with the platform's primary function. In addition to selling space for a single premium tier (advanced analytics), it may be possible to develop additional tiers of premium software to provide additional revenue sources, such as advanced reporting tools and developers who can access specific data interfaces associated with the platform. Corporate Social Responsibility (CSR) funding may help support projects that focus on issues regarding health, education, and sustainability.

We need to anticipate possible Project Risks and have fallback strategies to ensure the continued functioning of our systems in the event of an external data source or system outage. Financial Risks may include lack of funds and increased Operational Costs. Examples of Technology Risk include have outdated External Systems, Cyber Risks, and rapidly changing Technology Standards. Clear Agreements and Governance structures enable consistent collaboration, while Privacy and Data Protection issues help mitigate Reputational and Legal Risk through Open Communication and Privacy Conscious design choices. Organisational Risks arise when Local Authorities have varied Participations and vary in Technical Capability.

Reflecting on the implementation experience provides a number of important lessons learned. While the microservices design provided a great deal of flexibility to build functionality, it also required highly sophisticated deployment approaches. Data management processes related to privacy rules impacted how the data management teams made decisions on what data to collect. The lack of adequate documentation provided by many of the authorities led to the development of a standard wrapping services that joined several of the authorities' data sources. The feedback from the community led to greater accessibility and language support in the end-user interface design. In order to maintain the functions of the portal as a public service while maximising potential revenue opportunities, it was critical to develop a sustainable business model.

Building a Web Portal that combines Distributed Software and Service-oriented Architecture (SOA) for metropolitan-based Communities to take advantage of will lead to numerous opportunities for increasing Civic Engagement, Expediting Public Service Delivery, and strengthening the Linkages between all Community Members and their Local Municipality. This Portal will provide all Residents access to valuable information about their Community, expand the ability for Residents to see what their Government is doing for them, and prompt Residents to engage in Community-oriented Activities by providing them access to a centralized location where Community initiatives can be promoted. The establishment of this type of Portal will promote Social Cohesion, increase Government Responsiveness, and create a More Livable Environment for All. When Planning the Operational Components of the Portal, significant Infrastructure Developments must take place to allow for Growth and Scalability; appropriate Cybersecurity Measures that will ensure that any Sensitive Information that resides on the Portal, and Continual Monitoring and Evaluation of the Portal's Performance and Security must also occur, to ensure that the Portal has the Capacity to Meet Increasing Community Demands, accommodate changes in Technology and Legal Requirements, as well as changes in Community Needs.

Preserving accessibility and trust requires being sustainable (financially) through many different types of funding from Government Entities as well as collaborations with Community Organizations, responsible Advertising, and Optional additional services based upon Value Added principles. To demonstrate ways that financial sustainability will impact an individual's ability to have access to health & social services in his/her community is an example of an initial step in developing this capacity. This model of delivery provides a great opportunity for the development of adaptable architecture as well as comprehensive documentation; user-centric design and proactive risk management throughout the entire installation process and beyond allow these factors to play a role in transforming an online portal into an efficient, secure, agile, and ultimately effective way of improving individual and community health, empowering individuals, allowing for enhanced transparency between urban residents and local governments, and increasing the level of accountability between residents and government.